



Monitoring Report

CARBON OFFSET UNIT (CoU) PROJECT



Title: 136.5 MW Wind Power Projects in Brazil by Babilônia

Version: 2

Date :07/04/2025

First CoU Issuance Period: 5 years, 2 months

Date: 14/11/2018 to 31/12/2023



Monitoring Report (MR) CARBON OFFSET UNIT (CoU) PROJECT

Monitoring Report	
Title of the project activity	136.5 MW Wind Power Projects in Brazil by Babilônia
UCR Project Registration Number	387
Version	2
Completion date of the MR	07/04/2025
Monitoring period number and duration of this monitoring period	Monitoring Period Number: 1 Duration of this monitoring Period: (first and last days included (14/11/2018 to 31/12/2023))
Project participants	Project Owner: Babilônia Holding S.A. Project Aggregator: Kosher Climate India Private Limited
Host Party	Brazil
Applied methodologies and standardized baselines	Applied Baseline Methodology: ACM0002, version 21.0 – Consolidated methodology for grid-connected electricity generation from renewable source Standardized Methodology: Not Applicable
Sectoral scopes	01 Energy industries (Renewable/NonRenewable Sources)
Actual amount of GHG emission reductions for this monitoring period	2018: 32,513 CoUs (32,513 tCO _{2eq})
	2019: 264,814 CoUs (264,814 tCO _{2eq})
	2020: 22,203 CoUs (22,203 tCO _{2eq})
	2021: 97,486 CoUs (97,486 tCO _{2eq})
	2022: 113,256 CoUs (113,256 tCO _{2eq})
	2023: 0 CoUs (0 tCO _{2eq})
Total:	530,271 CoUs (530,271 tCO_{2eq})

SECTION A. Description of project activity

A.1. Purpose and general description of project activity >>

a) Purpose of the project activity and the measures taken for GHG emission reductions >>

The project **136.5 MW Wind Power Projects in Brazil by Babilônia** consists of five (5) project activities installed in Brazil, located in the state of Bahia, at the Village Orolândia. The promoter of the project is Babilônia Holding S.A., a company which has the full ownership of the project activity.

The details of the registered project are as follows:

Purpose of the project activity:

The purpose of the project activity is to generate electricity by harnessing the wind energy by the installation of Wind Turbine Generators (WTGs). The proposed project activity consists of five (5) project activities installed at the same location, clustered in a wind farm complex of total capacity 136.5 MW.

Project Activity	Power Plant Name	Village/State	Energy Source	Number of turbines	Capacity of each WTG	Installed capacity in MW	Annual generation in MWh/year	Commissioning date
1	BABILÔNIA I	Orolândia, BA	Wind	13	2.1	27.3	228,295	26/11/2018
2	BABILÔNIA II	Orolândia, BA	Wind	13	2.1	27.3	257,927	26/11/2018
3	BABILÔNIA III	Orolândia, BA	Wind	13	2.1	27.3	239,460	26/11/2018
4	BABILÔNIA IV	Orolândia, BA	Wind	13	2.1	27.3	267,644	26/11/2018
5	BABILÔNIA V	Orolândia, BA	Wind	13	2.1	27.3	364,683	14/11/2018

Emission reduction and impact of the project activity:

The project activity replaces anthropogenic emissions of greenhouse gases (GHGs) at approximately 530,271 tCO₂e over the entire crediting period, displacing an estimated average of 1,358,010 MWh from the generation-mix of power plants connected to the Brazilian grid. Project activity will mitigate the total GHG emission reductions of 530,271 tCO₂e over the entire crediting period, thereby contributing to climate change mitigation efforts.

The generated power from the project activities is being supplied to the Brazilian electricity grid. The project owners have signed a long-term power purchase agreement with CCEE (Câmara de Comercialização de Energia Elétrica). All five plants are commissioned and are currently operational and have been inter-connected to Morro do Chapéu II substation. Being a clean renewable energy source, wind power plants cause no negative impact on the environment. The project activity is thus promoting sustainable development, as defined by the United Nations, since economic advancement and progress have been fostered “(...) without compromising the ability of future generations to meet their own needs” (United Nations General Assembly, 1987, p. 43).

b) Brief description of the installed technology and equipment>>

Project activity consists of 5 installations – already installed, commissioned and under operation. The installed technologies convert the kinetic energy of the wind into mechanical energy and subsequently into electrical energy through Wind Turbine Generators (WTGs). This energy is further inverted into electricity.

The different components of the wind plant includes:

1. Wind Turbine Generators (WTGs),
2. Gear System & Coupling,
3. Generator,
4. Other relay and protection systems.

c) Relevant dates for the project activity (e.g. construction, commissioning, continued operation periods, etc.)>>

UCR Project ID or Date of Authorization: 387

Start Date of Crediting Period: 14/11/2018 to 31/12/2023

Project Commissioned:

Project Activity	Commissioning date
BABILÔNIA I	26/11/2018
BABILÔNIA II	
BABILÔNIA III	
BABILÔNIA IV	
BABILÔNIA V	14/11/2018

d) Total GHG emission reductions achieved or net anthropogenic GHG removals by sinks achieved in this monitoring period>>

The total GHG emission reductions achieved in this monitoring period is as follows:

Summary of the Project Activity and ERs Generated for the Monitoring Period	
Start date of this Monitoring Period	14/11/2018
Carbon credits claimed up to	31/12/2023
Total ERs generated (tCO _{2eq})	530,271 tCO _{2eq}
Leakage	0

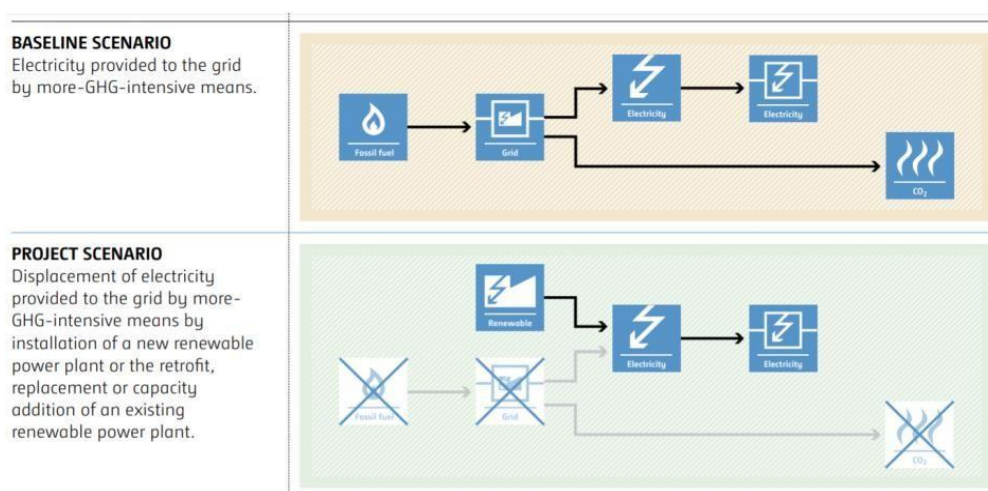
e) Baseline Scenario>>

The baseline scenario identified at the PCN stage of the project activity is:

As per the approved consolidated methodology ACM0002, version 21.0, if the project activity is the installation of a new grid-connected renewable power plant/unit, the baseline scenario is the following:

“Electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants”.

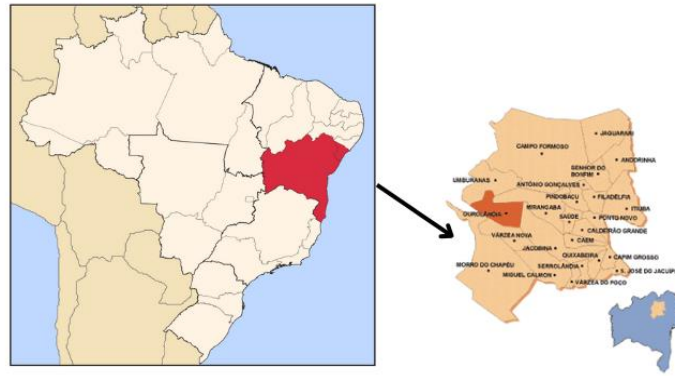
The project activity involves setting up a Wind power generation cluster to harness the power of wind, producing electricity and supplying it to the grid. In the absence of the project activity, the equivalent amount of power would have been supplied by the operation of grid-connected power plants. Therefore, the baseline scenario for the project activity is the equivalent amount of electricity generated from the Brazilian National grid.



A.2. Location of project activity>>>

Project activity consists of five (5) project activities installed at the same location, clustered in a wind farm complex of total capacity 136.5 MW:

Project Activity	Country	State	Village	Latitude	Longitude
BABILÔNIA I	Brazil	Bahia	Ourolândia	10°54'46.9" S	41°13'17.1" W
BABILÔNIA II	Brazil	Bahia	Ourolândia	10°56'22.9" S	41°13'44.0" W
BABILÔNIA III	Brazil	Bahia	Ourolândia	10°57'29.5" S	41°14'0.3" W
BABILÔNIA IV	Brazil	Bahia	Ourolândia	10°58'34.0" S	41°13'11.8" W
BABILÔNIA V	Brazil	Bahia	Ourolândia	11°0'22.9" S	41°13'37.7" W



A.3. Parties and project participants >>

Party (Host)	Participants
Brazil	Project Owner: Babilônia Holding S.A. Address: Rua José Gonçalves de Oliveira, 116 – 6th floor São Paulo, SP, Brazil Code 01453-050
India	Project Aggregator: KOSHER CLIMATE INDIA (P) LTD. Address: Zee Plaza, No. 1678, 27th Main Rd Bangalore, Karnataka, India Code 560102 Email: narendra@kosherclimate.com

A.4. References to methodologies and standardized baselines >>

SECTORAL SCOPE – 01, Energy industries (Renewable/Non-renewable sources)

TYPE – I - Renewable Energy Projects

CATEGORY – ACM0002: “Grid-connected electricity generation from renewable sources”, Version 21.0

A.5. Crediting period of project activity >>

Length of the crediting period corresponding to this monitoring period: 5 years, 2 months – 14/11/2018 - 31/12/2023

A.6. Contact information of responsible persons/entities >>

UCR ID – 387
Kosher Climate India Private Limited
Name: Narendra Kumar
Email ID – narendra@kosherclimate.com

SECTION B. Implementation of project activity

B.1. Description of implemented registered project activity >>

a) Provide information on the implementation status of the project activity during this monitoring period in accordance with UCR PCN>>

Project activity consists of 5 project activities and involves the installation of 65 Wind Turbine Generators (WTGs) of capacity 2.1MW each.

Technical specifications of the proposed components used for this project are given below:

WTG	
Design Data	
Rotor	GAMESA
Model	G114
Diameter	114 m
Cut-in wind speed	3 m/s
Cut-off wind speed	25 m/s
Generator	
Type	Double Fed Asyn
Nominal power	2.1 MW
Nominal Voltage	690 V

b) For the description of the installed technology(ies), technical process and equipment, include diagrams, where appropriate>>

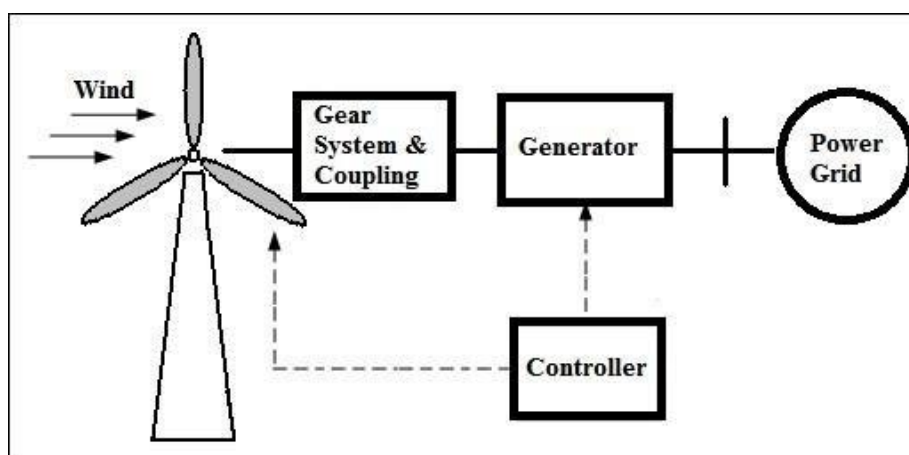


Figure: Installed Technologies, processes and equipment

B.2 Do no harm or Impact test of the project activity>>

As per the Environmental Impact Assessment (EIA) report conducted for the project, there is no harm associated with the project activity. Instead, there are social, environmental, economic and technological benefits which contribute to sustainable development, as described:

Social benefits:

- Employment opportunities created for the local workforce during project's construction and implementation phases;
- Employment opportunities to be created throughout the lifetime of the project activity;
- Development of rural and remote regions around the project activity.

Environmental benefits:

- Use of wind energy - a clean energy source - for generating electricity;
- Power generation with zero emission of GHG gases or specific pollutants like SO_x, NO_x, and SPM;
- Effort to minimize the dependence of the Brazilian energy matrix on fossil fuels;
- Minimum impact on land, water and soil at project surroundings.

Economic benefits:

- It fosters clean technology and clean energy investments in Brazil;
- It fosters the business development of local service providers in Brazil;
- Project activity can also provide new opportunities for industries and economic activities to be set in the area around the projects, developing rural and remote regions.

In addition to the social, environmental and economic benefits, the project activity also contributes to the sustainable development through supporting the local community and local economy thereby claiming SDGs 7, 8 and 13.

SDG 7: Ensure access to affordable, reliable, sustainable and modern energy for all.

Target 7.2: Increase global percentage of renewable energy.

KPI: Amount of renewable energy supplied to grid for consumption.

SDG 8: Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all.

Target 8.5: full employment and decent work with equal pay.

KPI: Average earning of females and male employees engaged in the project and segregated by age and persons with disabilities.

SDG 13: Take urgent action to combat climate change and its impacts

Target 13.2: Integrate climate change measures into national policies, strategies and planning

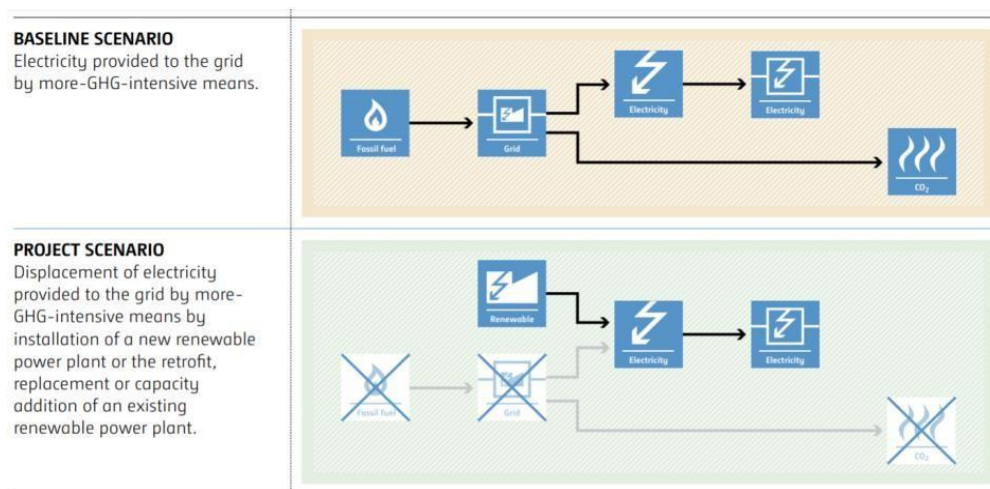
KPI: Amount of emission reductions achieved by project

B.3. Baseline Emissions>>

The baseline scenario identified at the PCN stage of the project activity is:

“Electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants”.

The project activity involves setting up a Wind power generation cluster to harness the power of wind, producing electricity and supplying it to the grid. In the absence of the project activity, the equivalent amount of power would have been supplied by the operation of grid-connected power plants. Therefore, the baseline scenario for the project activity is the equivalent amount of electricity generated from the Brazilian National grid.



B.4. Debundling>>

Project activity is not de-bundled component of any larger project.

SECTION C. Application of methodologies and standardized baselines

C.1. References to methodologies and standardized baselines >>

SECTORAL SCOPE – 01, Energy industries (Renewable/Non-renewable sources)

TYPE – I - Renewable Energy Projects

CATEGORY – ACM0002: “Grid-connected electricity generation from renewable sources”, Version 21.0

C.2. Applicability of methodologies and standardized baselines >>

Applicability Criteria	Applicability status
This methodology is applicable to grid-connected renewable power generation project activities that: (a) install Greenfield power plant; (b) involve a capacity addition to (an) existing plant(s); (c) involve a retrofit of (an) existing plant(s)/unit(s); (d) involve a rehabilitation of (an) existing plant(s)/unit(s); or (e) involve a replacement of (an) existing plant(s)/unit(s).	The project activities are newly installed green field wind energy-based electricity generation projects connected to the national grid. Therefore, it confirms to the said criteria.
In case the project activity involves the integration of a BESS, the methodology is applicable to grid-connected renewable energy power generation project activities that: (a) Integrate BESS with a Greenfield power plant; (b) Integrate a BESS together with implementing a capacity addition to (an) existing solar photovoltaic or solar power plant(s)/unit(s); (c) Integrate a BESS to (an) existing solar photovoltaic or solar power plant(s)/unit(s) without implementing any other changes to the existing plant(s); (d) Integrate a BESS together with implementing a retrofit of (an) existing solar photovoltaic or solar power plant(s)/unit(s).	The project activity involves the installation of a new grid connected renewable wind power projects and does not involve the integration of a Battery Energy Storage System (BESS). This condition is not applicable for the project activities.
The methodology is applicable under the following conditions: (a) Hydro power plant/unit with or without reservoir, solar power plant/unit, geothermal power plant/unit, solar power plant/unit, wave power plant/unit or tidal power plant/unit; (b) In the case of capacity additions, retrofits, rehabilitations or replacements (except for solar, solar, wave or tidal power capacity addition projects) the existing plant/unit started commercial operation prior to the start of a minimum historical reference period of five years, used for the calculation of baseline emissions and defined in the baseline emission section, and no capacity expansion, retrofit, or rehabilitation of the plant/unit has been undertaken between the start of this minimum historical reference period and the implementation of the project activity; (c) In case of Greenfield project activities applicable under paragraph 5 (a) above, the project participants shall demonstrate that the BESS was an integral part of the design of the renewable energy project activity (e.g., by referring to feasibility studies or investment decision documents); (d) The BESS should be charged with electricity generated from the associated renewable energy power plant(s). Only during	The project activities involve the installation of wind power plant/unit without BESS integration. Therefore, the said criteria is not applicable.

exigencies 2 may the BESS be charged with electricity from the grid or a fossil fuel electricity generator. In such cases, the corresponding GHG emissions shall be accounted for as project emissions following the requirements under section 5.4.4 below. The charging using the grid or using fossil fuel electricity generator should not amount to more than 2 per cent of the electricity generated by the project renewable energy plant during a monitoring period. During the time periods (e.g., week(s), months(s)) when the BESS consumes more than 2 per cent of the electricity for charging, the project participant shall not be entitled to issuance of the certified emission reductions for the concerned periods of the monitoring period.	
In case of hydro power plants, one of the following conditions shall apply: (a) The project activity is implemented in an existing single or multiple reservoirs, with no change in the volume of any of reservoirs; or (b) The project activity is implemented in an existing single or multiple reservoirs, where the volume of the reservoir(s) is increased and the power density calculated using equation (3) is greater than 4 W/m²; or (c) The project activity results in new single or multiple reservoirs and the power density calculate equation (3), is greater than 4 W/m². (d) The project activity is an integrated hydro power project involving multiple reservoirs, where the power density of any of the reservoirs, calculated using equation (3), is lower than or equal to 4 W/m², all of the following conditions shall apply. (i) The power density calculated using the total installed capacity of the integrated project, as per equation (4) is greater than 4W/m²; (ii) Water flow between reservoirs is not used by any other hydropower unit which is not a part of the project activity; (iii) Installed capacity of the power plant(s) with power density lower than or equal to 4 W/m² shall be: (a) Lower than or equal to 15 MW; and Less than 10% of the total installed capacity of integrated hydro power project.	The project activities involve the installation of a wind power plant/unit. Therefore, the said criteria is not applicable.
In the case of integrated hydro power projects, project proponent shall: (a) Demonstrate that water flow from upstream power plants/units spill directly to the downstream reservoir and that collectively constitute to the generation capacity of the integrated hydro power project; or Provide an analysis of the water balance covering the water fed to power units, with all possible combinations of reservoirs and without the construction of reservoirs. The purpose of water balance is to demonstrate the requirement of specific combination of reservoirs constructed under CDM project activity for the optimization of power output. This demonstration has to be carried out in the specific scenario of water availability indifferent seasons	The project activities involve the installation of a wind power plant/unit. Therefore, the said criteria is not applicable.

to optimize the water flow at the inlet of power units. Therefore, this water balance will take into account seasonal flows from river, tributaries (if any), and rainfall for minimum five years prior to implementation of CDM project activity.	
The methodology is not applicable to: (a) Project activities that involve switching from fossil fuels to renewable energy sources at the site of the project activity, since in this case the baseline may be the continued use of fossil fuels at the site. (b) Biomass fired power plants;	a) The project activities involve the installation of new wind power plant/unit. Which does not involve switching of grid-connected power plant. b) The project activities involve the installation of new wind power plant and not Biomass fired power plant. Therefore, the said criteria is not applicable.
In the case of retrofits, rehabilitations, replacements, or capacity additions, this methodology is only applicable if the most plausible baseline scenario, as a result of the identification of baseline scenario, is “the continuation of the current situation, that is to use the power generation equipment that was already in use prior to the implementation of the project activity and undertaking business as usual maintenance”.	The project activities involve the installation of new wind power plant/unit that does not involve retrofits, rehabilitations, replacements, or capacity additions. Therefore, the said criteria is not applicable

C.3 Applicability of double counting emission reductions >>

There is no double counting of emission reductions for the project activities due to the following reasons:

- Installations are uniquely identifiable based on its location coordinates;
- Project has dedicated commissioning certificates and connection points;
- Project is associated with energy meters which are dedicated to the consumption point for project developers;
- Projects are registered under [I-REC Standard](#) (IDs are presented next):

Project Activity	I-REC IDs
Babilônia I	EDP-001B
Babilônia II	EOLVWIND001
Babilônia III	EOLVWIND002
Babilônia IV	EOLVWIND003
Babilônia V	EOLVWIND004

C.4. Project boundary, sources and greenhouse gases (GHGs)>>

The project boundary includes the physical, geographical sites of the wind cluster, sub-stations, grid and all power plants connected to the national grid. The project activity will evacuate power to the Brazilian National Grid. Therefore, the entire Brazilian national grid and all connected power plants have been considered in the project boundary for the project activities.

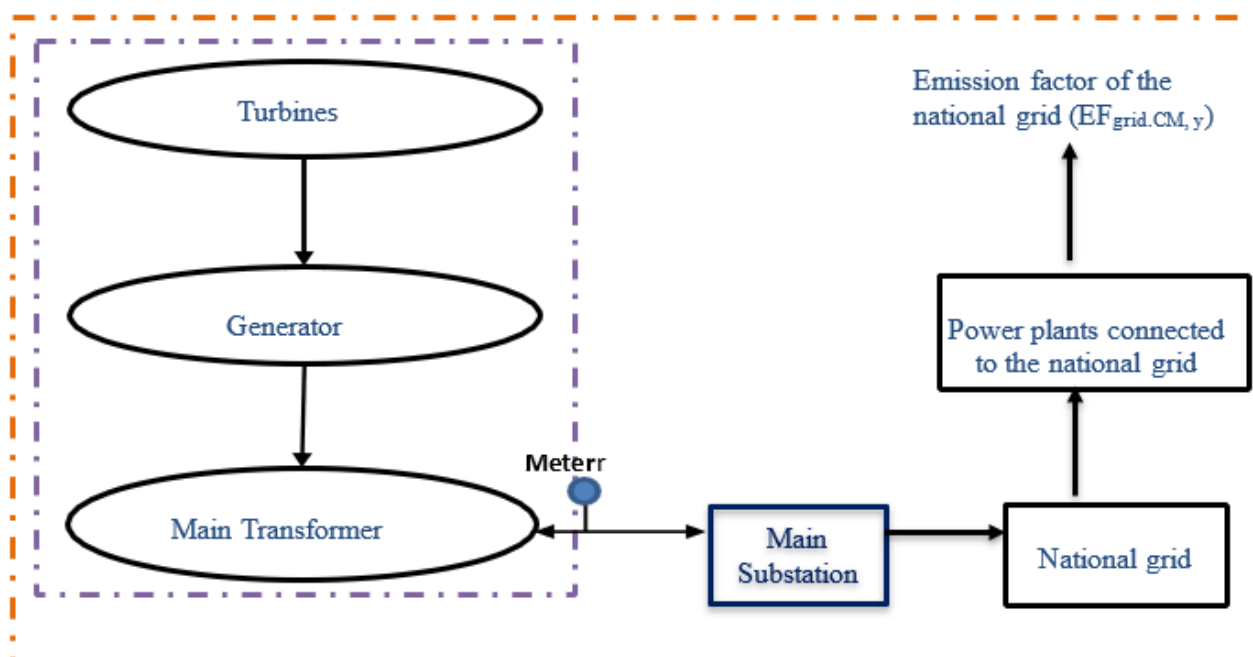


Figure: Project Boundary

The project does not involve any other emissions sources not foreseen by the methodologies. The table below provides an overview of the emissions sources included or excluded from the project boundary for determination of baseline and project emissions.

Source		GHG	Included?	Justification/Explanation
Baseline	Grid Connected Electricity Generation	CO ₂	Yes	Main Emission Source
		CH ₄	No	Minor Emission source
		N ₂ O	No	Minor Emission source
Project Activity	Greenfield Wind Power Project activity	CO ₂	No	Project activity does not emit CO ₂
		CH ₄	No	Project activity does not emit CH ₄
		N ₂ O	No	Project activity does not emit N ₂ O

C.5. Establishment and description of baseline scenario (UCR Protocol) >>

As per the methodology ACM0002, Version 21.0, if the project activity is the installation of a new grid-connected renewable power plant/unit, the baseline scenario is the following:

“The baseline scenario is that the electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources into the grid.”

The project activity involves setting up new wind power plants to harness the power of wind energy and inject electricity into the Brazilian regional grid. In the absence of the project activity,

the equivalent amount of power would have been generated by the operation and/or insertion of more- GHG-intensive grid-connected power plants. Hence, the baseline for the project activity is the equivalent amount of power produced at the Brazilian grid.

Net GHG Emission Reductions and Removals:

$$ER_y = BE_y - PE_y$$

Where,

ER_y = Emission reductions in year y (t CO₂e/yr)

BE_y =Baseline emissions in year y (t CO₂/yr)

PE_y =Project emissions in year y (t CO₂/yr)

Baseline Emissions

As per the approved consolidated Methodology ACM0002 version 21.0, Baseline emissions include only CO₂ emissions from electricity generation in fossil fuel fired power plants that are displaced due to the project activity. The methodology assumes that all project electricity generation above baseline levels would have been generated by existing grid-connected power plants and the addition of new grid- connected power plants. The baseline emissions are to be calculated as follows:

$$BE_y = EG_{PJ,y} \times EF_{grid,CM,y}$$

Where,

BE_y = Baseline emissions in year y (t CO₂/yr)

$EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh/yr)

$EF_{grid,CM,y}$ = Combined margin CO₂ emission factor for grid connected power generation in year y calculated using the latest version of the “Tool to calculate the emission factor for an electricity system” (t CO₂/MWh)

As per para 49 of ACM0002, version 21.0, when the project activity is installation of Greenfield power plant, then:

$$EG_{PJ,y} = EG_{facility,y}$$

Where,

$EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh/yr)

$EG_{facility,y}$ = Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh/yr)

The combined margin of the Brazil grid used for the project activity is as follows:

Parameter	Value	Nomenclature	Source
-----------	-------	--------------	--------

EF _{grid,CM,y}	As given below	Combined margin CO ₂ emission factor for the project electricity system in year y.	Calculated as the weighted average of the operating margin (0.75) & build margin (0.25) values, sourced from Ministry of Science and Technology “CO ₂ emission factors for electricity generation in the National Interconnected System of Brazil.
EF _{grid,OM,y}	As given below	Operating margin CO ₂ emission factor for the project electricity system in year y.	Calculated from the monthly average emission, sourced from Ministry of Science and Technology “CO ₂ emission factors for electricity generation in the National Interconnected System of Brazil.
EF _{grid,BM,y}	As given below	Build margin CO ₂ emission factor for the project electricity system in year y.	The built margin value has been sourced from Ministry of Science and Technology “CO ₂ emission factors for electricity generation in the National Interconnected System of Brazil.

Year	Operating Margin Emission Factor	Build Margin Emission Factor	Combined Margin Emission Factor
2018	0.5390	0.137	0.4385
2019	0.5181	0.1020	0.4140
2020	0.4539	0.0979	0.3649
2021	0.5985	0.0540	0.4623
2022	0.3963	0.0270	0.3039
2023	0.3785	0.0467	0.2956

Project Emissions:

As the project activity consists of the installation of a new grid-connected wind power plant and does not involve any project emissions from fossil fuel, operation of dry, flash steam or binary geothermal power plants, and from water reservoirs of hydro power plants. Therefore, project emissions are:

$$PE_y = 0.$$

Where,

PE_y = Project emissions in year y (t CO₂e/yr)

Leakage Emissions:

No other leakage emissions are considered. The emissions potentially arising due to activities such as power plant construction and upstream emissions from fossil fuel use (e.g. extraction, processing, transport etc.) are neglected.

Hence Emission reductions will be calculated as per the below equation:

$$ER_y = BE_y = EG_{PJ,y} \times EF_{grid,CM,y}$$

The actual emission reduction achieved during the crediting period has been demonstrated below as a part of first monitoring and verification.

The following table summarizes the yearly actual net generation and its respective yearly actual emission reduction, as per the given yearly emission factor:

Crediting period	Net Generation free from I-REC (MWh/year)	Emission Factor (tCO ₂ /MWh)	Emission Reduction (tCO ₂ /year)
------------------	---	---	---

14-11-2018 to 31-12-2018	74,230.90	0.44	32,513.13
01-01-2019 to 31-12-2019	639,540.21	0.41	264,813.61
01-01-2020 to 31-12-2020	60,847.25	0.36	22,203.16
01-01-2021 to 31-12-2021	210,827.85	0.46	97,485.78
01-01-2022 to 31-12-2022	372,563.93	0.30	113,256
01-01-2023 to 31-12-2023	0	0.30	0
Total	1,358,010		530,271

C.6. Prior History>>

Project Activities are registered under [I-REC Standard](#) (IDs are presented next):

Project Activity	I-REC IDs
Babilônia I	EDP-001B
Babilônia II	EOLVWIND001
Babilônia III	EOLVWIND002
Babilônia IV	EOLVWIND003
Babilônia V	EOLVWIND004

C.7. Monitoring period number and duration>>

First Issuance Period: 14/11/2018 – 31/12/2023.

C.8. Changes to start date of crediting period >>

There is no change in the start date of crediting period.

C.9. Permanent changes from PCN monitoring plan, applied methodology or applied standardized baseline >>

The present document is the Monitoring Report version 1.0, which presents no changes from the PCN monitoring plan, applied methodology or applied standardized baseline.

C.10. Monitoring plan>>

Data and Parameters available at validation (ex-ante values):

Data/Parameter	EF _{grid,CM,y}
Data unit	tCO ₂ /MWh
Description	Combined Margin CO ₂ emission factor in year y of Brazilian Grid

Source of data	Designated National Authority (DNA) “Ministry of Science and Technology” CO2 emission factors for electricity generation in the National Interconnected System of Brazil ¹ ”						
Value(s) applied	Year	2018	2019	2020	2021	2022	2023
	Value (t CO ₂)	0.438	0.4141	0.3649	0.4624	0.3040	0.2956
Measurement methods and procedures	-						
Monitoring frequency	Ex-ante fixed parameter						
Purpose of data	Calculation of Emission Factor of the grid						

Data/Parameter	EF _{grid,OM,y}						
Data unit	tCO ₂ /MWh						
Description	Operating Margin CO ₂ emission factor in year y of Brazilian Grid						
Source of data	Designated National Authority (DNA) “Ministry of Science and Technology” CO ₂ emission factors for electricity generation in the National Interconnected System of Brazil ² ”						
Value(s) applied							
	Year	2018	2019	2020	2021	2022	2023
	Value (t CO ₂)	0.539	0.5181	0.4539	0.5985	0.3963	0.3785
Measurement methods and procedures	-						
Monitoring frequency	Ex-ante fixed parameter						
Purpose of data	Calculation of Emission Factor of the grid						

Data/Parameter	EF _{grid,BM,y}						
Data unit	tCO ₂ /MWh						
Description	Build Margin CO ₂ emission factor in year y of Brazilian Grid						
Source of data	Designated National Authority (DNA) “Ministry of Science and Technology” CO ₂ emission factors for electricity generation in the National Interconnected System of Brazil ³ ”						
Value(s) applied							
	Year	2018	2019	2020	2021	2022	2023
	Value (t CO ₂)	0.137	0.1020	0.0979	0.054	0.027	0.0467
Measurement methods and procedures	-						
Monitoring frequency	Ex-ante fixed parameter						
Purpose of data	Calculation of Emission Factor of the grid						

Data and Parameters to be monitored (ex-post monitoring values):

Data / Parameter:	EG _{facility,y}
Data unit:	MWh/Year
Description:	Quantity of net electricity generation supplied by the project (Wind) plant/unit to the grid in year y
Source of data:	Monthly Generation Report

¹ <https://www.gov.br/mcti/pt-br/acompanhe-o-mcti/sirene/dados-e-ferramentas/fatores-de-emissao>

² <https://www.gov.br/mcti/pt-br/acompanhe-o-mcti/sirene/dados-e-ferramentas/fatores-de-emissao>

³ <https://www.gov.br/mcti/pt-br/acompanhe-o-mcti/sirene/dados-e-ferramentas/fatores-de-emissao>

Measurement procedures (if any):

	Main Meter			Back-Up Meter		
Type of meter	Bidirectional Meter			Bidirectional Meter		
Location of meter	Power plant site			Power plant site		
Accuracy of meter	0.2s			0.2s		
Serial number of meters	Proj. Act.	Series nº		Proj. Act.	Series nº	
	1	MW-1707A005-02		1	MW-1706A781-02	
	2	MW-1706B374-02		2	MW-1707A024-02	
	3	MW-1706B451-02		3	MW-1706A865-02	
	4	MW-1706A757-02		4	MW-1706B476-02	
	5	MW-1706B470-02		5	MW-1706A693-02	
Calibration frequency	Once every 5 years			Once every 5 years		
Date of Calibration/ validity	2017:			2017:		
	Main Meter nº	Calibration date	Calibration validity	Back-up Meter nº	Calibration date	Calibration validity
	MW-1707A005-02	30/08/2017	30/08/2022	MW-1706A781-02	15/09/2017	15/09/2022
	MW-1706B374-02	31/08/2017	31/10/2022	MW-1707A024-02	31/08/2017	31/10/2022
	MW-1706B451-02	31/08/2017	31/10/2022	MW-1706A865-02	15/09/2017	15/09/2022
	MW-1706A757-02	30/08/2017	30/08/2022	MW-1706B476-02	31/08/2017	31/10/2022
	MW-1706B470-02	31/08/2017	31/10/2022	MW-1706A693-02	30/08/2017	30/08/2022
	2024:			2024:		
	Main Meter nº	Calibration date	Calibration validity	Main Meter nº	Calibration date	Calibration validity
	MW-1707A005-02	09/01/2024	09/01/2029	MW-1706A781-02	09/01/2024	09/01/2029
	MW-1706B374-02	09/01/2024	09/01/2029	MW-1707A024-02	09/01/2024	09/01/2029
	MW-1706B451-02	09/01/2024	09/01/2029	MW-1706A865-02	09/01/2024	09/01/2029
	MW-1706A757-02	10/01/2024	10/01/2029	MW-1706B476-02	10/01/2024	10/01/2029

		MW-1706 B470 -02	10/0 1/20 24	10/0 1/20 29		-02	MW-1706 A693 -02	10/0 1/20 24	10/0 1/20 29
	Reference No. of Calibration Certificate	2017:			2017:				
		Proj. Act.	Certificate n°		Proj. Act.	Certificate n°			
		1	9809/Z-17		1	9667/Z-17			
2		9840/Z-17		2	9869/Z-17				
3		9835/Z-17		3	9974/Z-17				
4		9814/Z-17		4	9830/Z-17				
5		9826/Z-17		5	9813/Z-17				
2024:			2024:						
Proj. Act		Certificate n°		Proj. Act	Certificate n°				
1		CC-0853-24		1	CC-0854-24				
2	CC-0855-24		2	CC-0856-24					
3	CC-0857-24		3	CC-0858-24					
4	CC-0859-24		4	CC-0860-24					
5	CC-0861-24		5	CC-0862-24					
Calibration Status	Calibrated			Calibrated					
Continuous									
13,58,010 MWh									
The backup meters are installed alongside the main meters on the same panel, providing the same information for current and voltage, meeting the same technical standard. Qualified organizations, adhering to national standards and regulations, conduct meter calibration to ensure accurate measurements.									
The calibration of plant-end meters will be carried out once in 5 years or whenever abnormal difference/inconsistency is observed between main meters and back-up meters.									
There is calibration delay for the monitoring period from September 2022 to December 2023. The error factor has been applied in net export values for delay period as meters were not calibrated as per the calibration frequency. As per VVS requirement: error factor of "±0.2%" should be applicable for both export & import i.e. the measured values. Hence being conservative and to account for the error for both export & import, a cumulative error of “-0.2% on export and +0.2% for import values” has been applied for delay period.									
Any comment:									
-									